

SkyBridge Airways is a full-service international airline replacing several legacy operations tools with a unified Flight Operations and Maintenance Control Platform. The target state combines dispatch planning, crew assignment, aircraft routing, maintenance event tracking, minimum equipment list deferral workflows, turnaround status, and operational messaging into one enterprise workflow.

The platform is used by dispatchers in the airline operations center, maintenance controllers, crew schedulers, station managers, safety investigators, and a small set of privileged reliability-engineering and vendor-support personnel. Flight crews interact with selected data through the airline's Electronic Flight Bag (EFB) synchronization service and receive dispatch releases, notices, and aircraft-specific status packages before departure. Airport station teams and contracted ground handlers use a restricted turnaround portal to update gate events, fueling completion, baggage loading milestones, de-icing status, and pushback readiness.

The architecture is intentionally hybrid. Core identity, privileged access management, records retention, and several system-of-record databases remain on premises in airline data centers. Workflow applications, API mediation, rules processing, and analytics workloads run in a primary public cloud tenant with a warm standby region. The airline also depends on external providers for weather, flight-plan optimization, airport decision-making feeds, maintenance data, vendor support, and crew operational messaging.

Several workflows are safety- and compliance-critical. Only authorized dispatchers can finalize flight releases, and those releases must incorporate weather, aircraft status, turnaround readiness, crew legality, and maintenance deferral state. Maintenance control can create and close engineering events, but only licensed controllers can approve selected return-to-service decisions. Crew schedulers can recover disruptions by reassigning pairings during irregular operations. These urgent override paths are exactly where airline leadership expects attackers to focus because they can bypass normal review steps when the network is already under pressure.

A major concern is the boundary between operational truth and operational status updates. Ground handlers can submit turnaround events and exceptions through partner APIs. Maintenance vendors can upload health-monitoring alerts and recommended actions. ACARS and EFB channels can deliver aircraft-tail-specific information to crews. Auditors have repeatedly flagged that break-glass diagnostics, vendor support sessions, and emergency maintenance workflows are not consistently threat-modeled, especially when multiple systems disagree about aircraft status and operators rely on exception handling to keep flights moving.